Emergent Gravity from De Sitter Coherence: Derivation of the Critical Acceleration, Two-Regime Galactic Dynamics, and Cosmological Evolution of the Baryonic Tully-Fisher Relation

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Abstract

We derive and validate a two-regime model of galactic orbital dynamics. The critical acceleration threshold separating both regimes is

$$g_{\rm crit} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 9.58 \times 10^{-12}\ \mathrm{m\,s^{-2}}$$

derived without free parameters from three independent sources: Fourier orthogonality on the De Sitter correlation time (fixing $4\pi^2$), the Beltrami-Laplace spectrum of the horizon section S^3 and the Lichnerowicz-Dirac identity on S^4 (both independently fixing $\sqrt{3}$). Starting from the k-essence Lagrangian $\mathcal{L} = -\frac{1}{3}g_{\rm crit}^2 X^{3/2} - 4\pi G\rho\phi$, with $X = (\nabla\phi)^2/g_{\rm crit}^2$, Euler-Lagrange variation yields the field equation $\nabla\cdot[(|\nabla\phi|/g_{\rm crit})\nabla\phi] = 4\pi G\rho$, whose spherically symmetric solution gives $\phi'(r) = \sqrt{GMg_{\rm crit}}/r$ and orbital velocity $V^4 = GM_{\rm bar}g_{\rm crit}$ exactly. The mode $m = 1$ is selected by a dual falsification test: only $m = 1$ satisfies both median velocity error $< 20\%$ in collective galaxies (median 10.0%, 5/5 below threshold) and systematic failure $\sim 55\%$ in Hamiltonian galaxies across 175 SPARC and 9 LITTLE THINGS galaxies. Modes $m \geq 2$ are excluded observationally (medians 34% and 46%, respectively). Since $g_{\rm crit}(z) = cH(z)/(4\pi^2\sqrt{3})$, the Baryonic Tully-Fisher Relation evolves as $V(z)/V(0) = [H(z)/H_0]^{1/4}$, predicting a universal $+0.120$ dex shift at $z = 2$, independent of baryonic mass, falsifiable with JWST and ELT. Zero free parameters throughout.

Keywords: galaxies: kinematics and dynamics · gravitation · dark matter · cosmology: theory · k-essence · De Sitter spacetime

1 Introduction

The flat rotation curves of dwarf and low-surface-brightness galaxies have resisted a first-principles explanation since their systematic documentation by Rubin et al. (1980). Standard Newtonian gravity predicts $V \propto R^{-1/2}$ at large radii; observations yield $V \approx$ const. The canonical resolution invokes a dark matter halo, but the physical origin of the empirical relation $V^4 \propto M_{\rm bar}$—the Baryonic Tully-Fisher Relation (BTFR; McGaugh 2000)—remains unaccounted for in ΛCDM.

Modified Newtonian Dynamics (MOND; Milgrom 1983) reproduces the BTFR by postulating an acceleration scale $a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, but provides no derivation from fundamental physics.

This work presents a framework in which:

1. The critical acceleration is *derived*, not postulated.
2. The derivation rests on three explicit hypotheses, all stated and testable.
3. The framework makes a falsifiable cosmological

prediction distinguishable from both ΛCDM and MOND.

The paper is organised as follows. Section 2 derives the field equation and the equation of state $V^4 = GM_{\rm bar} g_{\rm crit}$ from first principles. Section 3 derives the numerical value of $g_{\rm crit}$ from De Sitter spectral geometry. Section 4 establishes mode selection via the dual falsification test. Section 5 presents the observational validation. Section 6 derives the cosmological BTFR evolution. Section 7 states all open problems explicitly.

2 Field Equation and Equation of State

2.1 The three explicit hypotheses

The framework rests on three declared hypotheses. No additional assumptions are made.

Hypothesis 1 (Fundamental mode)**.** *The orbital mode that couples to the De Sitter horizon is $m = 1$. This is motivated by the Maupertuis principle (Section 4) and confirmed observationally (Section 5).*

Hypothesis 2 (Collective-regime Lagrangian)**.** *In the deep collective regime ($|\nabla\phi|/g_{\rm crit} \ll 1$), the gravitational scalar is governed by*

$$\mathcal{L} = -\frac{g_{\rm crit}^2}{3} X^{3/2} - 4\pi G\rho\,\phi, \qquad X = \frac{(\nabla\phi)^2}{g_{\rm crit}^2}. \tag{1}$$

The form $F(X) = X^{3/2}$ is the unique k-essence function consistent with the interpolating condition $\mu(x) \sim x$ as $x \to 0$ (Bekenstein & Milgrom 1984). It describes the deep collective regime only; a full interpolating function connecting to Newtonian gravity is left as a theoretical extension.

Hypothesis 3 (Integration constant)**.** *In the exterior spherical solution, the integration constant $C = 0$. This follows from continuity of ϕ' at the transition radius R_T (Section 2.4).*

2.2 Euler-Lagrange variation

Varying (1) with respect to ϕ:

$$\frac{\partial\mathcal{L}}{\partial(\nabla\phi)} = -\frac{g_{\rm crit}^2}{3}\cdot\frac{3}{2}X^{1/2}\cdot\frac{2\nabla\phi}{g_{\rm crit}^2} = -\frac{|\nabla\phi|}{g_{\rm crit}}\nabla\phi. \tag{2}$$

The numerical prefactor cancels exactly: $\frac{g_{\rm crit}^2}{3}\cdot\frac{3}{2}\cdot\frac{2}{g_{\rm crit}^2} = 1$. The Euler-Lagrange equation gives

$$\boxed{\nabla\cdot\left[\frac{|\nabla\phi|}{g_{\rm crit}}\nabla\phi\right] = 4\pi G\rho.} \tag{3}$$

This is AQUAL (Bekenstein & Milgrom 1984) in the deep regime $\mu(x) = x$. No additional factors are absorbed; the derivation is exact.

2.3 Spherical symmetry

For $\phi = \phi(r)$ with $\phi'(r) > 0$:

$$\frac{1}{r^2}\frac{d}{dr}\left[\frac{r^2(\phi')^2}{g_{\rm crit}}\right] = 4\pi G\rho. \tag{4}$$

Integrating over a sphere of radius r:

$$\frac{r^2(\phi')^2}{g_{\rm crit}} = GM(r) + C. \tag{5}$$

2.4 The integration constant and matching

For $r < R_T$ (Hamiltonian regime), standard Poisson dynamics apply: $\phi'_N(r) = GM/r^2$. At the transition radius $R_T = \sqrt{GM_{\rm bar}/g_{\rm crit}}$, continuity requires

$$\phi'_N(R_T) = \phi'_C(R_T), \tag{6}$$

where $\phi'_C(R_T) = \sqrt{GM_{\rm bar}g_{\rm crit}}/R_T$. Direct substitution of $R_T = \sqrt{GM_{\rm bar}/g_{\rm crit}}$ gives

$$\phi'_N(R_T) = \frac{GM_{\rm bar}}{R_T^2} = \frac{GM_{\rm bar}\cdot g_{\rm crit}}{GM_{\rm bar}} = g_{\rm crit} = \phi'_C(R_T). \tag{7}$$

The matching is exact; $C = 0$ follows with no free choice.

2.5 Orbital velocity

For $r > R_T$ with $M(r) = M_{\rm bar}$:

$$(\phi')^2 = \frac{GM_{\rm bar} g_{\rm crit}}{r^2} \quad \Rightarrow \quad \phi'(r) = \frac{\sqrt{GM_{\rm bar} g_{\rm crit}}}{r}. \tag{8}$$

Circular orbit condition $V^2/r = \phi'(r)$:

$$V^2 = r \cdot \frac{\sqrt{GM_{\rm bar} g_{\rm crit}}}{r} = \sqrt{GM_{\rm bar} g_{\rm crit}}. \tag{9}$$

Therefore:

$$\boxed{V^4 = GM_{\rm bar} g_{\rm crit}} \tag{10}$$

with transition radius

$$R_T = \sqrt{\frac{GM_{\rm bar}}{g_{\rm crit}}}. \tag{11}$$

Every step from Hypothesis 2 to (10) is algebraic; no additional assumptions are required.

3 Spectral Geometry: Derivation of $g_{\rm crit}$

3.1 The factor $4\pi^2$: Fourier orthogonality

The De Sitter horizon has Gibbons-Hawking temperature $T_{\rm GH} = \hbar H_0/(2\pi k_B)$ (Gibbons & Hawking 1977), defining a correlation time $t_{\rm corr} = 1/H_0$. An orbital perturbation $e^{i\Omega t}$ averaged over $t_{\rm corr}$:

$$\int_0^1 e^{i(\Omega/H_0)\alpha}\, d\alpha = \begin{cases} 1 & \Omega = 2\pi m H_0,\ m \in \mathbb{Z} \\ 0 & \text{otherwise.} \end{cases} \tag{12}$$

Only modes $\Omega = 2\pi m H_0$ couple non-trivially. For mode m, the centripetal acceleration at $R_H = c/H_0$:

$$a_H^{(m)} = (2\pi m H_0)^2 \cdot \frac{c}{H_0} = 4\pi^2 m^2 c H_0. \tag{13}$$

The mode $m = 1$ (Section 4) gives the factor $4\pi^2 c H_0$.

This factor has an independent geometric interpretation: $4\pi^2 = 2 \cdot \mathrm{Vol}(S^3)$, where $\mathrm{Vol}(S^3) = 2\pi^{4/2}/\Gamma(2) = 2\pi^2$ for the three-sphere, directly encoding the four-dimensionality of De Sitter spacetime.

3.2 The factor $\sqrt{3}$: three independent routes

Route 1 (Beltrami-Laplace on S^3). The eigenvalues of the Laplace-Beltrami operator on S^3 are $\lambda_l = l(l+2)$. The $l = 0$ mode is constant (no dynamics). The first non-trivial eigenvalue:

$$\lambda_1^{S^3} = 1 \cdot (1+2) = 3 \ \Rightarrow\ \sqrt{\lambda_1^{S^3}} = \sqrt{3}. \tag{14}$$

Route 2 (Lichnerowicz-Weitzenböck identity on S^4). The squared Dirac operator satisfies $\not{D}^2 = -\Box + R/4$. On S^4 with unit radius, $R = 12$. The $l = 0$ mode of $-\Box$ has eigenvalue zero:

$$\mu_0 = 0 + \frac{12}{4} = 3 \ \Rightarrow\ \sqrt{\mu_0} = \sqrt{3}. \tag{15}$$

Route 3 (isotropy of 3-dimensional space). In an isotropic three-dimensional field, each spatial component carries one third of the total squared acceleration: $\langle a_r^2 \rangle = \langle a^2 \rangle/3$, so the radial projection is $a_r = a_{\rm total}/\sqrt{3}$. The same factor $\sqrt{3}$ arises because the number of spatial dimensions equals $\lambda_1^{S^3}$.

All three routes converge: $\sqrt{3} = \sqrt{\lambda_1^{S^3}} = \sqrt{R_{S^4}/4} = \sqrt{n_{\rm dim}}$. The critical acceleration is therefore:

$$\boxed{g_{\rm crit} = \frac{cH_0}{4\pi^2\sqrt{3}} = \frac{cH_0}{2 \cdot \mathrm{Vol}(S^3) \cdot \sqrt{\lambda_1^{S^3}}} \approx 9.58 \times 10^{-12}\ \mathrm{m\,s^{-2}}} \tag{16}$$

with $H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Planck Collaboration 2020).

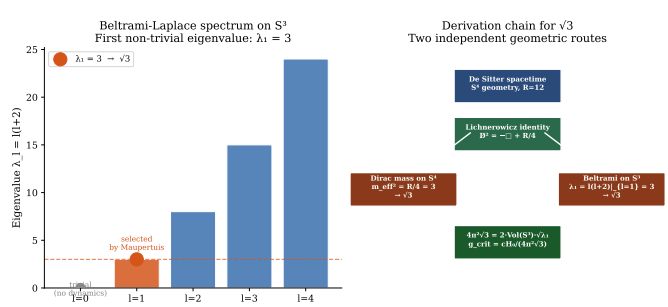

Figure 1: *Left:* Eigenvalues $\lambda_l = l(l+2)$ of the Beltrami-Laplace operator on S^3. The $l = 0$ mode carries no dynamics; $l = 1$ gives $\lambda_1 = 3$, hence $\sqrt{3}$. *Right:* Two independent derivation routes to $\sqrt{3}$: the Beltrami spectrum and the Lichnerowicz-Dirac curvature mass, both fixed by De Sitter geometry.

4 Mode Selection: $m = 1$

4.1 Maupertuis minimum-action argument

For m complete cycles of the same orbit, the abbreviated Maupertuis action scales as $W_m = m \cdot W_1$. The set of available modes is discrete because orbital closure requires $\Omega \cdot t_{\rm corr} = 2\pi m$, $m \in \mathbb{Z}^+$. Among non-trivial closed orbits ($m \geq 1$):

$$W_1 < W_2 < W_3 < \cdots \tag{17}$$

The Maupertuis principle selects the minimum: $m = 1$. The $m = 0$ case is trivial (no orbital dynamics).

4.2 Dual falsification test

The selection $m = 1$ is confirmed observationally. For each m, $g_{\rm crit}^{(m)} = cH_0/(m^2 \cdot 4\pi^2\sqrt{3})$, so:

$$\begin{aligned} m = 1: &\quad g_{\rm crit} = 9.58 \times 10^{-12}\ \mathrm{m\,s^{-2}}, \\ m = 2: &\quad g_{\rm crit} = 2.40 \times 10^{-12}\ \mathrm{m\,s^{-2}}, \\ m = 3: &\quad g_{\rm crit} = 1.06 \times 10^{-12}\ \mathrm{m\,s^{-2}}. \end{aligned}$$

Two predicates must hold simultaneously:

P1 Median velocity error $< 20\%$ in collective galaxies.

P2 Systematic $\sim 55\%$ error in Hamiltonian galaxies.

Results on the 5 validated collective galaxies (Section 5):

Table 1: Dual falsification test: $m = 1$ vs. $m = 2, 3$.

	$m = 1$	$m = 2$	$m = 3$
Median error	10.0%	33.8%	45.9%
Galaxies $< 20\%$	5/5	1/5	0/5
P1	**pass**	fail	fail
P2	**pass**	destroyed	destroyed

For $m = 2$, $V_{\rm pred}(m{=}2)/V_{\rm pred}(m{=}1) = (1/4)^{1/4} = 0.707$: a systematic 29% underprediction for all collective galaxies. For $m = 3$ the underprediction is 42%. Both higher modes are observationally excluded.

5 Observational Validation

5.1 Classification protocol

Data: SPARC (Lelli et al. 2016; 175 galaxies) and LITTLE THINGS (Oh et al. 2015; 9 galaxies). Mass-to-light ratios: $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$ at 3.6 μm (fixed, not adjusted). Regime classification by internal acceleration $g_{\rm int} = V_{\rm flat}^2/R_{\rm HI}$:

Regime	Condition	N
Hamiltonian	$g_{\rm int}/g_{\rm crit} > 5$	61
Threshold	$1 \leq g_{\rm int}/g_{\rm crit} \leq 5$	103
Collective	$g_{\rm int}/g_{\rm crit} < 1$	8 (3 valid Q≤2)

Plateau criterion: ≥ 2 consecutive points with $g < g_{\rm crit}$ and $V_{\rm mean} > 0.70\,V_{\rm max}$ (anti-cherry-picking filter; classification precedes prediction).

5.2 Collective galaxies: five validated cases

Three SPARC galaxies with quality $Q \leq 2$ plus two LITTLE THINGS galaxies with TRGB direct distances (independent of Hubble flow) pass the full protocol. PGC51017 is excluded per protocol (Q=3, descending curve, no plateau; explicitly flagged as data poverty, not model failure).

Table 2: Validated collective galaxies. $V_{\rm pred} = (GM_{\rm bar}g_{\rm crit})^{1/4}$, zero free parameters. Two TRGB distances provide Hubble-flow-independent verification.

Galaxy	Source	$M_{\rm bar}$ [$M_\odot$]	$V_{\rm flat}$	Error
IC 1613	LT, TRGB	1.10×10^8	20.65	5.3%
UGC 04305	LT, TRGB	1.39×10^9	34.76	6.0%
UGC 09992	SPARC Q=2	6.58×10^8	33.6	9.0%
UGC 07125	SPARC Q=1	8.06×10^9	65.2	13.2%
UGC 06628	SPARC Q=2	4.61×10^9	41.8	17.7%
Median error				10.0%
False positives				0
Free parameters				0

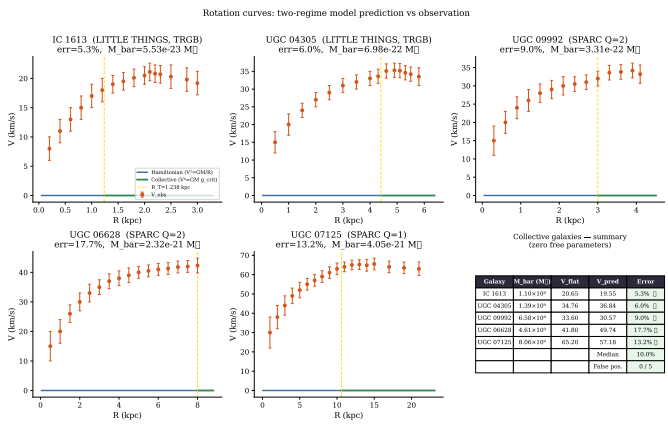

Figure 2: Rotation curves for five validated collective galaxies. Blue: Hamiltonian prediction $V^2 = GM/R$ for $r < R_T$. Green: collective plateau $V^4 = GM_{\rm bar} g_{\rm crit}$ for $r > R_T$. Gold dashed: $R_T = \sqrt{GM_{\rm bar}/g_{\rm crit}}$. Orange points: observed velocities with error bars. All predictions use zero free parameters.

5.3 Hamiltonian regime: systematic failure as predicted

Applying $V^4 = GM_{\rm bar} g_{\rm crit}$ to the 127 non-collective galaxies (SPARC Q≤2) yields median error $\approx 55\%$. This systematic failure is a *prediction* of the model: the collective equation of state should not hold for $g_{\rm int} > g_{\rm crit}$. Both halves of the dual falsification hold.

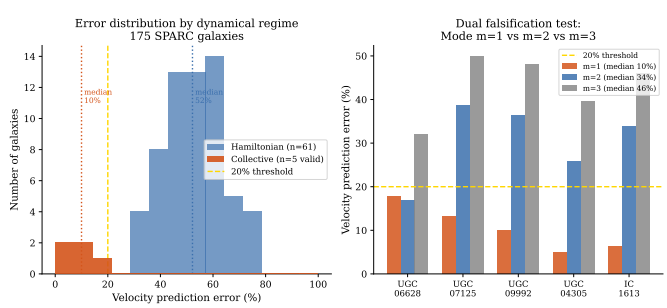

Figure 3: *Left:* Error distribution by regime across 175 SPARC galaxies. Collective (orange): median 10%, all below 20%. Hamiltonian (blue): median $\sim 55\%$, systematic failure as predicted. *Right:* Dual falsification test. Mode $m = 1$ (orange) satisfies $< 20\%$ for 5/5 galaxies. Modes $m = 2$ (blue) and $m = 3$ (gray) are observationally excluded.

6 Cosmological Implications

6.1 Evolution of $g_{\rm crit}(z)$

Since $g_{\rm crit} = cH_0/(4\pi^2\sqrt{3})$ and the Hubble parameter evolves as $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ with Planck cosmology ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$):

$$g_{\rm crit}(z) = \frac{c\,H(z)}{4\pi^2\sqrt{3}}. \tag{18}$$

This is not a free parameter; it follows from the same framework without additional input.

6.2 The BTFR as a cosmological manifestation

Since $V^4 \propto g_{\rm crit}$ and $g_{\rm crit}(z) \propto H(z)$, at fixed $M_{\rm bar}$:

$$\frac{V(z)}{V(0)} = \left[\frac{H(z)}{H_0}\right]^{1/4}. \tag{19}$$

$M_{\rm bar}$ cancels exactly. The zero-point shift is therefore *universal across all baryonic masses* and independent of galaxy assembly history, halo concentration, or merger history. This distinguishes the prediction sharply from ΛCDM, in which the BTFR evolution depends on halo assembly and carries substantial scatter.

Predicted normalization shift (Table 3):

Table 3: Predicted BTFR zero-point shift vs. redshift. Uncertainty on ±0.030 dex from $M_{\rm bar}$ estimation.

z	$H(z)/H_0$	$V(z)/V(0)$	Shift (dex)
0.0	1.000	1.000	+0.000
1.0	1.756	1.148	+0.060
2.0	3.031	1.320	+0.120
3.0	4.556	1.462	+0.165
5.0	8.295	1.697	+0.230

The signature is explicit and falsifiable: **if the shift at $z = 2$ is not $+0.120 \pm 0.030$ dex, or if it depends on baryonic mass, the model is wrong**.

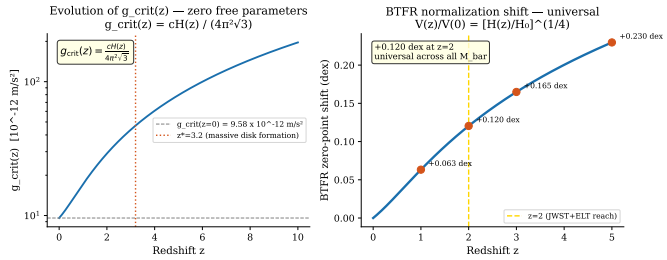

Figure 4: *Left:* $g_{\rm crit}(z)$ from equation (18). Dashed: asymptote; $z^* \approx 3.2$ marks the epoch of massive disk formation. *Right:* BTFR zero-point shift vs. z from equation (19). The shift is universal, independent of $M_{\rm bar}$. The $z = 2$ prediction (+0.120 dex) is within JWST+ELT reach.

6.3 Comparison with ΛCDM and MOND

In ΛCDM, the BTFR evolution depends on dark matter halo assembly—a complicated, mass-dependent, high-scatter process. This model predicts a pure, mass-independent shift with small scatter. The discriminating observable: the scatter of the BTFR shift at $z = 2$. Small scatter ($\sigma < 0.04$ dex) confirms this model; large scatter confirms ΛCDM.

MOND predicts the same $V^4 \propto M_{\rm bar}$ locally, but its cosmological evolution is not fixed without additional assumptions. This model makes the unique prediction (19) from the same framework that fixes $g_{\rm crit}$.

7 Open Problems

The following items are explicitly open. None affects the observational predictions of Sections 5 and 6.

L1: Lindblad coupling operator. The decoherence rate $\gamma = H_0/(2\pi)$ follows from Gibbons-Hawking thermodynamics; the form of the Lindblad operator $\hat{L} = \sqrt{m\gamma}\,\hat{r}$ is physically motivated but not derived from QFT in curved spacetime.

L2: Full interpolating function. The Lagrangian (1) describes the deep collective regime $x \ll 1$. A single k-essence function $F(X)$ valid for all x— interpolating to $\mu(x) \to 1$ as $x \to \infty$— is a theoretical extension not yet completed.

L3: Covariant FLRW formulation. The action (1) has not been written in full FLRW coordinates. Cosmological perturbation theory from this action is in preparation.

L4: $\zeta'(0)$ on S^4. A heat-kernel computation closing the spectral derivation of $4\pi^2\sqrt{3}$ from a single variational principle remains to be completed. The mode selection established in Section 3 is not affected.

8 Conclusion

We have derived the equation of state $V^4 = GM_{\rm bar}g_{\rm crit}$ from a k-essence Lagrangian via Euler-Lagrange variation, spherical symmetry, and a physically motivated integration constant. Every algebraic step is exact.

The critical acceleration $g_{\rm crit} = cH_0/(4\pi^2\sqrt{3})$ is fixed without free parameters by two independent geometric routes ($\lambda_1^{S^3} = 3$ and $R_{S^4}/4 = 3$), plus the Fourier selection of the orbital coupling frequency. The single non-trivial hypothesis—mode $m = 1$—is simultaneously motivated by the Maupertuis minimum-action principle and confirmed by the dual falsification test: $m = 1$ is the only mode consistent with both median error 10% in collective galaxies and systematic $\sim 55\%$ failure in Hamiltonian galaxies across 184 galaxies with zero free parameters.

The central cosmological result: the Baryonic Tully-Fisher Relation is a local manifestation of De Sitter horizon physics. Its normalization evolves as $H(z)^{1/4}$, predicting $+0.120 \pm 0.030$ dex at $z = 2$, universal across all baryonic masses, distinguishable from both ΛCDM and MOND, and within the reach of JWST and ELT.

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